



US against the moon & mars?

15 years and an investment of \$50 billion, still most space ships in the US are not ready to launch. <https://digit.in/july19-32>



Astronauts return

Russian and North American astronauts return safely to Earth following a launch accident. <https://digit.in/july19-33>



The Pragyan Rover

Credit: ISRO

CHANDRAYAAN 2

India's follow up mission to the Moon

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On June 13, Dr K Sivan, the Chairman of ISRO, met with the press to announce some additional details about the next milestone mission for the Indian space program, Chandrayaan 2. The mission has been scheduled for a July 15 launch, at 2:51 AM in the morning.

The mission has been in the planning stage for an incredibly long duration. The Chandrayaan 2 mission was approved in 2008, even before Chandrayaan was launched. Initially targeted for the first quarter of 2018, ISRO deliberately delayed the launch of Chandrayaan-2 to make sure that they got the mission right, without rushing through the preparation.

MISSION PROFILE

There are three components of the mission. The lander is stacked above the orbiter within the rocket. The orbiter weighs 2,379 kg, and is capable of communicating with the lander as well as the Indian Deep Space Network (IDSN) facilities near Bengaluru. The lander is named Vikram in honour of Dr Vikram Sarabhai, the founder of the Indian space program. The lander can communicate with the rover, the orbiter, and the IDSN facility. The lander has a planned mission duration of 1 lunar day, or 14 Earth days. The lander weighs 1,471 kg, and has the rover nested within it. The rover is named Pragyan (which means wisdom in Sanskrit), weighs 27 kg and is the lightest component on the mission. The rover generates electricity using a solar panel, has six wheels, and can only communicate with the lander. All the three components, stacked together, form a composite body that is located in the heat shield or payload fairing of the GSLV MK III rocket, India's most pow-

erful launch vehicle, and one that has been entirely designed and fabricated within the country.

The launch will take place from the Sriharikota High Altitude Range (SHAR), India's launch complex. Initially, the composite payload will be deployed into an Earth parking 170x40400 km orbit. Over the period of the next 16 days, the altitude of the composite body will be increased five times, using on board thrusters. On the 16th day, ISRO will execute a maneuver called trans lunar injection, where the spacecraft jumps from an Earth orbit to the Moon. The Chandrayaan 2 mission will take five days to make the hop from the Earth to the Moon. At the end of the period, the thrusters will be fired again to inject the spacecraft into an orbit around the Moon. Over the course of the next 27 days, the on board propulsion systems will fire continuously, to reduce the altitude of the lunar orbit, down to a circular orbit of 100x100 km. The orbiter will be used to survey the landing site near